

LQR Control of an Inverted Pendulum on a Cart

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EMCH 792: Learning-Based Controls

We reuse the nonlinear Simulink model from Project 1 and add a horizontal cart force. Since designing a controller directly from nonlinear equations is complex, we linearize them near upright. First, we use $\sin(\theta) \approx \theta$ and $\cos(\theta) \approx 1$. Second, we discard the product $\theta\dot{\theta}^2$, which does not contribute to the first-order model. This gives $\dot{z} = Az + BF$ for $z = (x, \dot{x}, \theta, \dot{\theta})^T$. Appendix A gives the derivation.

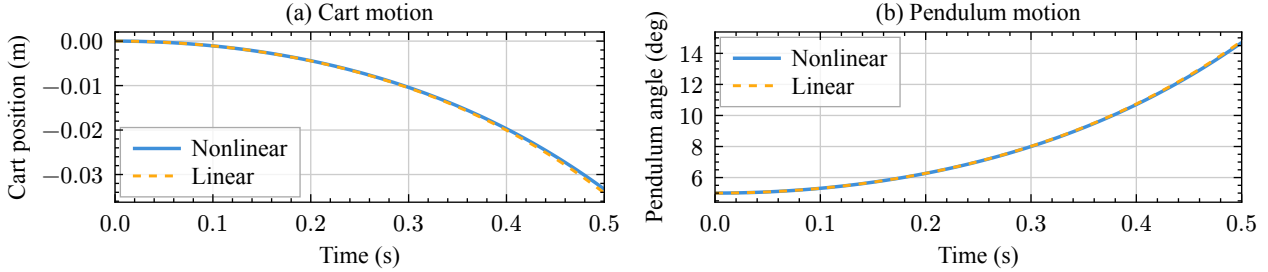


Figure 1: Linear and nonlinear responses to the same unforced 5° release over 0.5 s.

Figure 1 shows that the models respond closely near upright. Their largest differences are 0.094° and 0.87 mm. This supports using the linear model for local controller design. Before designing that controller, we form $C = [B, AB, A^2B, A^3B]$. MATLAB finds rank four, equal to the number of states, so the system is controllable [1].

Since we know the model and have all four states, we choose a linear-quadratic regulator (LQR). A proportional-integral-derivative (PID) controller can instead be tuned from measured response when a reliable model is unavailable [2]. Here, LQR lets us assign a cost to each state error and to the force command [3]. We select weights of 4 for cart position, 2 for cart velocity, 300 for angle, and 10 for angular velocity, collected in $Q = \text{diag}(4, 2, 300, 10)$. The force penalty is $R = 0.5$. These are design choices that emphasize balance while discouraging unnecessary motion and force. Appendix B gives the design and essential MATLAB functions.

MATLAB calculates $u = -Kz$ and confirms that all poles of $A - BK$ have negative real parts. We also simulate the linear closed loop and verify that its states return to zero. We then connect this controller to the nonlinear Simulink model in Appendix E. The actuator limits u to ± 10 N, and a separate disturbance d is added afterward, so the cart receives $F = u + d$.

To test the controller, we run two categories of 30 s simulations. First, we release the pendulum from 5, 10, 20, 30, 45, and 60° to test the linearization and actuator limits. Second, we start upright and apply random cart forces with bounds of $\pm 2.5, 5, 7.5$, and 10 N. Each force is an independent uniform draw held for 0.1 s. The same seeded sequence is scaled for each bound. Appendix C explains this generation method and its scope. Appendix D gives the full metrics and actuator histories.

RESULTS.

For the initial-angle tests, Figure 2(a,c) shows cart position and pendulum angle over 30 s. Panels (b,d) show 0–6 s details. The 5, 10, and 20° cases recover; the tested 30, 45, and 60° cases fail. The transition from successful recovery to failure to recover therefore lies between the tested 20° and 30° releases.

For the random-force tests, Figure 3(a,c,e) shows cart position, pendulum angle, and applied forces over 30 s. Panels (b,d,f) show 0–3 s details. The controller maintains balance at the tested 2.5, 5, and 7.5 N bounds, but loses balance at 10 N. At 2.5 N, the RMS angle is 0.69° and the peak is 1.79°.

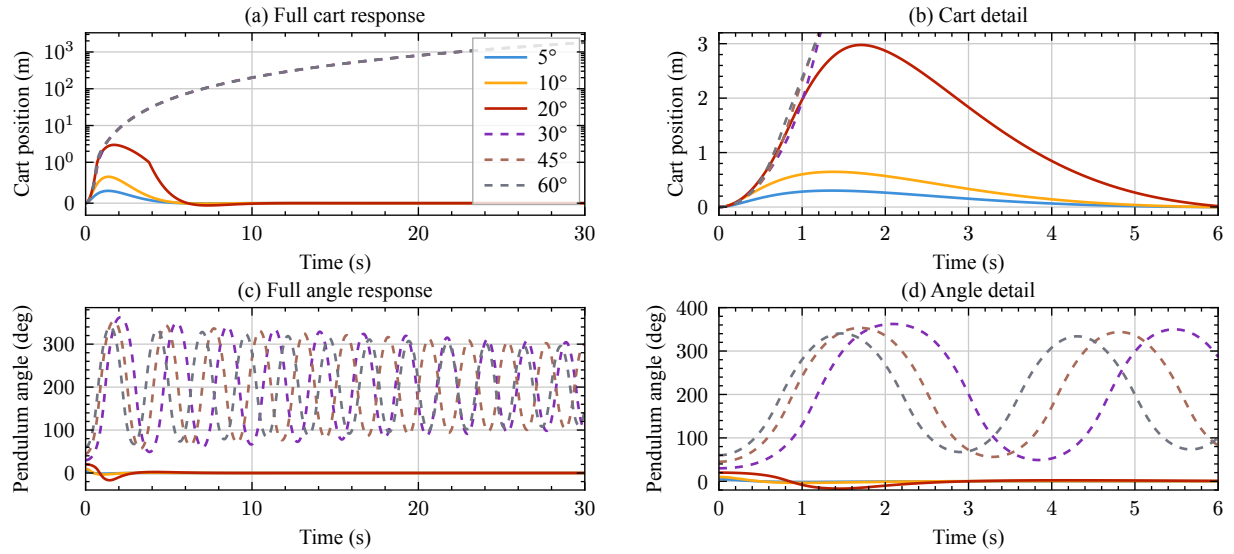


Figure 2: Initial-angle sweep. Dashed traces indicate initial angles where the controller failed to stabilize. Panels (b,d) show 0–6 s. Panel (a) uses a symmetric log scale, linear within ± 1 m.

DISCUSSION.

The smaller tested inputs recover or remain balanced. Failed releases stay at the actuator limit, and the 10 N disturbance also causes sustained saturation. Even successful recovery at 20° needs 2.98 m of cart travel, so a physical track could constrain performance. The sweeps identify an operating range for this gain and force limit. The random-force result applies to the recorded sequence described in Appendix C.

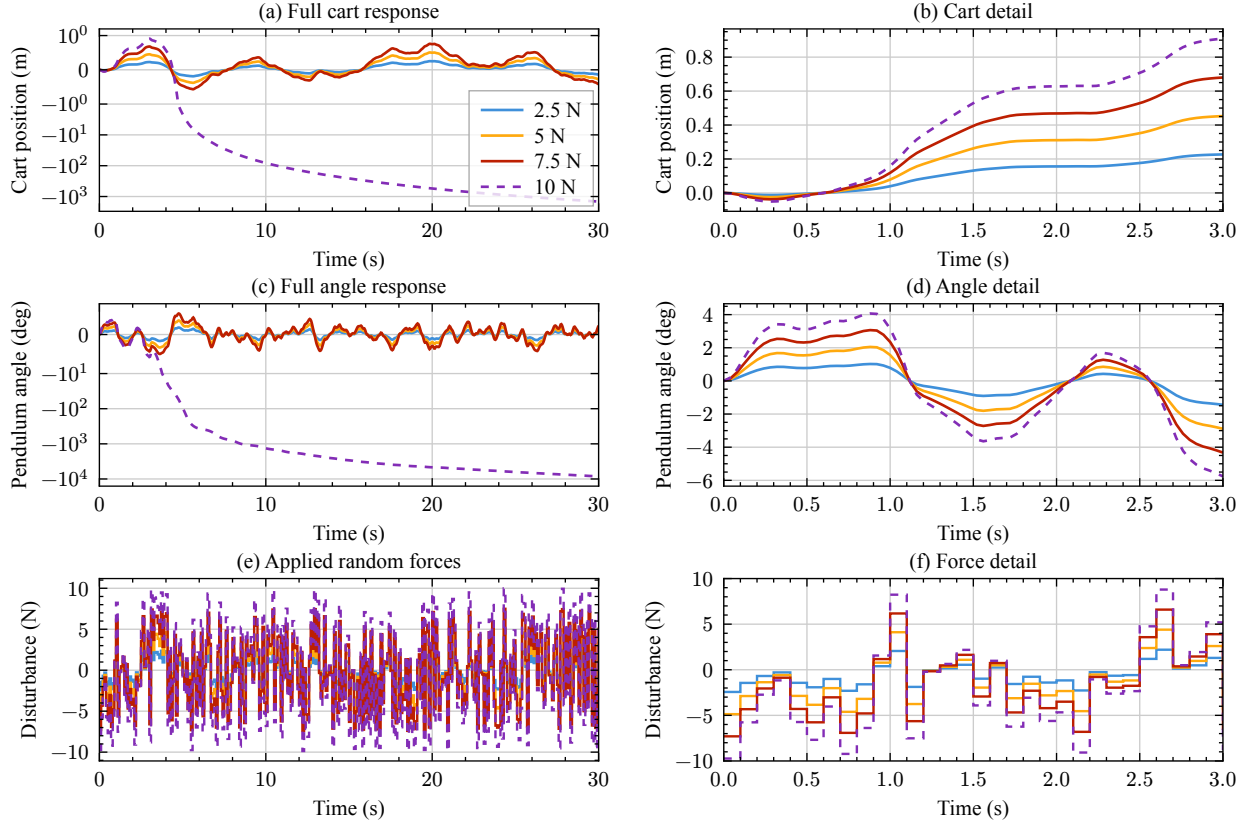


Figure 3: Random-force sweep. Every bound uses the same sequence of force draws; the 10 N case is dashed. Panels (b,d,f) show 0–3 s. Panels (a,c) use symmetric log scales, linear within ± 1 m and $\pm 10^\circ$, respectively.

Appendix A. Nonlinear model, linearization, and model comparisons

We reuse the nonlinear model from Project 1, along with its mechanical assumptions. The cart translates horizontally on a frictionless track. A rigid, massless rod carries a point mass, and a viscous rotational damper acts at the pivot. Motion is planar, gravity is constant, and the track has no end stops. Project 2 adds the horizontal cart force F to the original equations,

$$(M + m)\ddot{x} + m\ell \cos(\theta)\ddot{\theta} - m\ell \sin(\theta)\dot{\theta}^2 = F,$$

$$m\ell \cos(\theta)\ddot{x} + m\ell^2\ddot{\theta} + c_\theta\dot{\theta} - mg\ell \sin(\theta) = 0.$$

Here x is cart displacement, θ is the pendulum angle measured from upright, and dots indicate time derivatives. Positive F acts in the positive x direction. The symbols M , m , ℓ , c_θ , and g denote cart mass, pendulum point mass, rod length, pivot damping coefficient, and gravitational acceleration, respectively. Table 1 gives the values retained from Project 1, using the mass symbols in the equations and MATLAB implementation.

Table 1: Mechanical parameters reused from Project 1.

Parameter	Meaning	Value	Unit
M	Cart mass	2.0	kg
m	Pendulum point mass	0.5	kg
ℓ	Rod length	1.0	m
c_θ	Pivot damping	0.01	N m s/rad
g	Gravity	9.81	m/s ²

Because the controller is intended to hold the pendulum upright, we expand about the stationary equilibrium $\theta = \dot{\theta} = \ddot{\theta} = F = 0$. Cart position can be chosen as $x = 0$ because the equations do not depend on absolute position. For angles in radians, the Taylor expansions give

$$\sin(\theta) = \theta - \frac{\theta^3}{6} + \dots, \quad \cos(\theta) = 1 - \frac{\theta^2}{2} + \dots$$

Consequently, $\sin(\theta) \approx \theta$ and $\cos(\theta) \approx 1$ retain the first-order behavior near upright. These approximations become accurate as the angle approaches zero. For example, at 5° , replacing $\sin(\theta)$ by θ introduces approximately 0.13% relative error, and replacing $\cos(\theta)$ by one introduces approximately 0.38% relative error.

We also simplify the centrifugal term, $m\ell \sin(\theta)\dot{\theta}^2$, which represents the horizontal effect of the pendulum's rotational motion on the cart. Near upright, replacing $\sin(\theta)$ by θ makes this term approximately $m\ell\theta\dot{\theta}^2$. It contains three factors that approach zero near the stationary equilibrium. One is the angle θ , and the other two are the angular velocity $\dot{\theta}$ multiplied by itself.

A first-order model retains terms proportional to a single small deviation, such as θ , $\dot{\theta}$, or F . The centrifugal term instead multiplies these deviations together. For example, reducing both the angle and angular velocity by a factor of ten reduces $\theta\dot{\theta}^2$ by a factor of one thousand, while the retained gravity and damping terms decrease by only a factor of ten. Its effect therefore becomes much smaller than the retained terms as the motion approaches upright and rest. We discard it in this local linear model. This step assumes both a small angle and a small angular velocity; a small angle alone does not justify discarding it during rapid rotation. With this term removed and the trigonometric approximations substituted, the equations become

$$(M + m)\ddot{x} + m\ell\ddot{\theta} = F,$$

$$m\ell\ddot{x} + m\ell^2\ddot{\theta} + c_\theta\dot{\theta} - mg\ell\theta = 0.$$

With the linearized equations established, we separate the accelerations to obtain the state-space model. Divide the second linearized equation by $m\ell$ and rearrange it,

$$\ell\ddot{\theta} = -\ddot{x} - \frac{c_\theta}{m\ell}\dot{\theta} + g\theta.$$

Substituting this expression into the first equation eliminates $\ddot{\theta}$,

$$(M + m)\ddot{x} + m\left(-\ddot{x} - \frac{c_\theta}{m\ell}\dot{\theta} + g\theta\right) = F,$$

$$M\ddot{x} = F + \frac{c_\theta}{\ell}\dot{\theta} - mg\theta.$$

Dividing by M and substituting the result back into the angular equation gives

$$\ddot{x} = \frac{F}{M} - \frac{mg}{M}\theta + \frac{c_\theta}{M\ell}\dot{\theta},$$

$$\ddot{\theta} = -\frac{F}{M\ell} + \frac{(M + m)g}{M\ell}\theta - \frac{c_{\theta(M+m)}}{Mm\ell^2}\dot{\theta}.$$

These two second-order equations become four first-order equations by defining

$$z = \begin{pmatrix} z_1 \\ z_2 \\ z_3 \\ z_4 \end{pmatrix} = \begin{pmatrix} x \\ \dot{x} \\ \theta \\ \dot{\theta} \end{pmatrix}, \quad \dot{z} = \begin{pmatrix} z_2 \\ -\frac{mg}{M}z_3 + \frac{c_\theta}{M\ell}z_4 + \frac{F}{M} \\ z_4 \\ \frac{(M+m)g}{M\ell}z_3 - \frac{c_{\theta(M+m)}}{Mm\ell^2}z_4 - \frac{F}{M\ell} \end{pmatrix}.$$

Collecting the state coefficients and force coefficients gives $\dot{z} = Az + BF$. With the parameters in Table 1, the matrices are

$$A = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & -2.4525 & 0.005 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 12.2625 & -0.025 \end{pmatrix}, \quad B = \begin{pmatrix} 0 \\ 0.5 \\ 0 \\ -0.5 \end{pmatrix}.$$

Before designing the controller, MATLAB forms the controllability matrix $\mathcal{C} = [B, AB, A^2B, A^3B]$ and finds $\text{rank}(\mathcal{C}) = 4$. The rank equals the four-state dimension, so the cart force can control all four states in this linear model.

To check whether this linear model captures the nonlinear motion near upright, we compare both models under identical initial conditions and inputs, without feedback or actuator limits. Each runs for 1 s using MATLAB's ode45 solver, relative tolerance 10^{-10} , absolute tolerance 10^{-12} , and a maximum step of 0.002 s. The nonlinear model uses the Simulink plant's acceleration function.

Because uncontrolled motion grows away from upright, we compare errors at 0.1, 0.25, and 0.5 s rather than 10 or 20 s. Relative to the nonlinear response, the percentage errors are

$$e_{x(t)} = 100 \frac{|x_{\text{lin}(t)} - x_{\text{nonlin}(t)}|}{|x_{\text{nonlin}(t)}|}, \quad e_{\theta(t)} = 100 \frac{|\theta_{\text{lin}(t)} - \theta_{\text{nonlin}(t)}|}{|\theta_{\text{nonlin}(t)}|}.$$

Percentage errors are undefined at a zero reference value. All reference values at the selected times are nonzero.

First, we release both models from 2, 5, 10, 15, 20, 25, and 30°, with zero cart position, zero velocities, and zero applied force. Figure 4 compares the motion, and Table 2 reports the differences at the selected times.

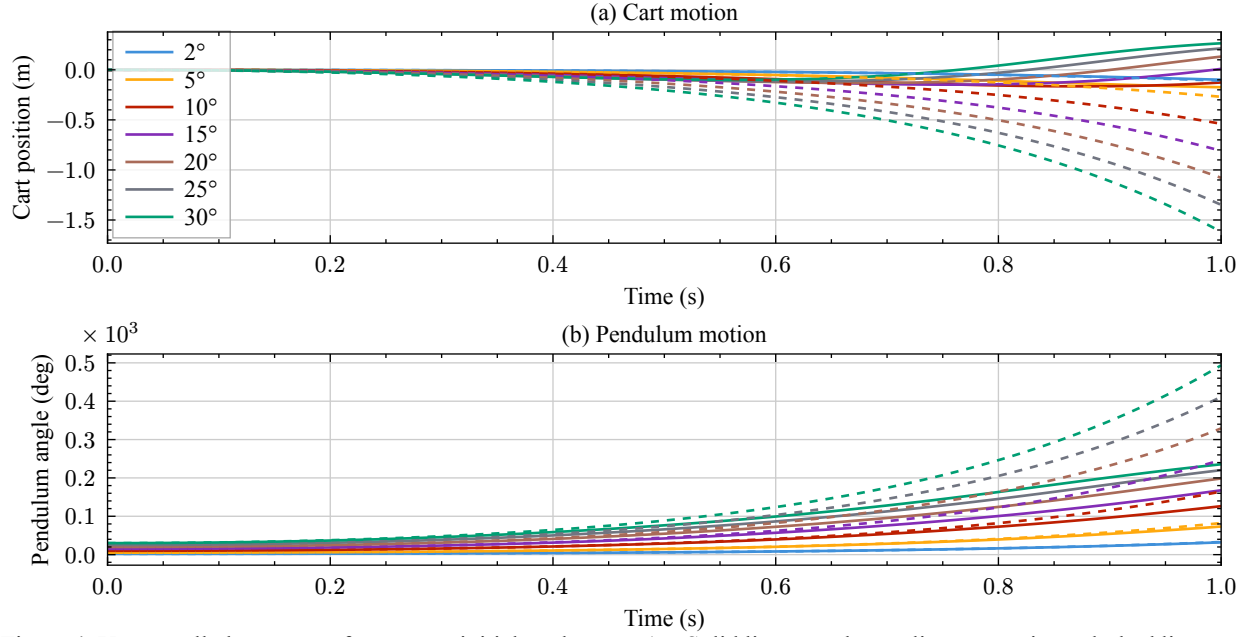


Figure 4: Uncontrolled responses from seven initial angles over 1 s. Solid lines use the nonlinear equations; dashed lines use the upright linearization. Matching colors indicate matching initial angles.

Table 2: Percentage model errors following unforced releases, relative to the nonlinear response at each comparison time.

Initial angle (deg)	0.1 s		0.25 s		0.5 s	
	e_x (%)	e_θ (%)	e_x (%)	e_θ (%)	e_x (%)	e_θ (%)
2	0.1178	0.0031	0.1559	0.0193	0.4149	0.1034
5	0.7384	0.0192	0.9773	0.1202	2.5991	0.6377
10	2.9833	0.0761	3.9505	0.474	10.5033	2.441
15	6.827	0.1681	9.049	1.0429	24.1582	5.149
20	12.4341	0.2916	16.5106	1.7992	44.6605	8.4763
25	20.0601	0.442	26.7172	2.7101	74.2447	12.1901
30	30.0821	0.614	40.2567	3.7416	117.1998	16.1266

At 0.5 s, the 2° release has an angle error of 0.103% and a cart-position error of 0.415%. At 30°, those errors increase to 16.13% and 117.2%, respectively. The increasing differences show why the linear model is appropriate near upright and why larger releases require verification with the nonlinear plant.

Second, we start both models upright and at rest and apply a constant horizontal force from $t = 0$. The six force levels are 0.5, 1, 2, 4, 8, and 16 N. Figure 5 shows the responses, and Table 3 reports the differences. These are prescribed inputs for model validation, separate from the later controller tests with a 10 N actuator limit.

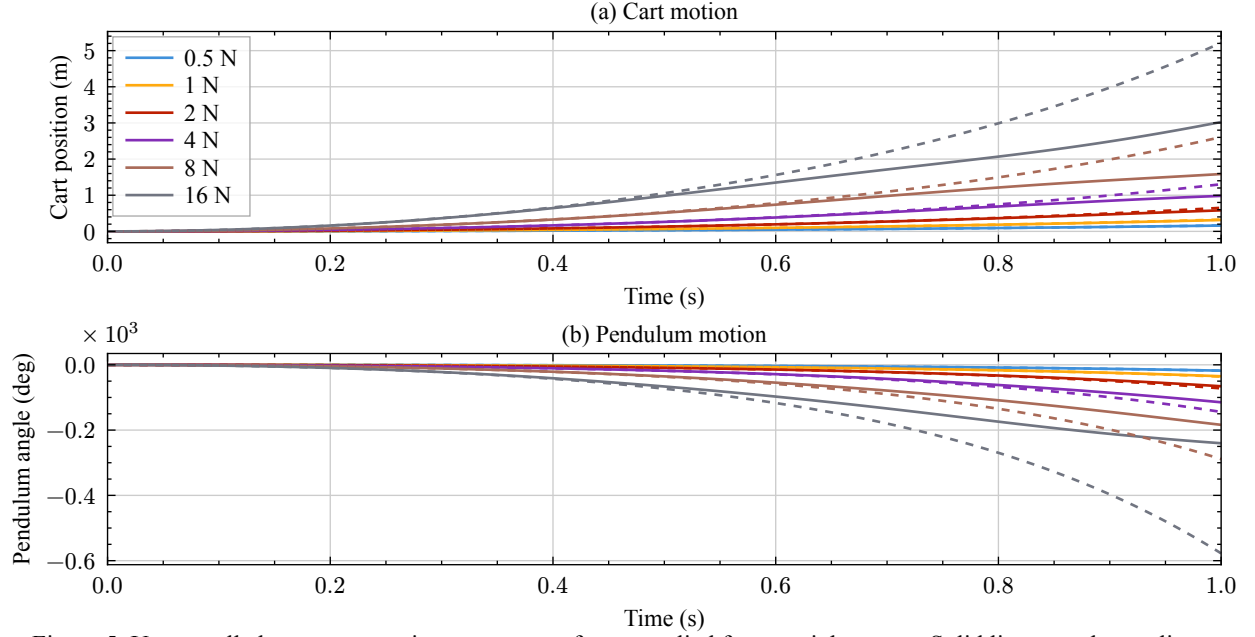


Figure 5: Uncontrolled responses to six constant cart forces applied from upright at rest. Solid lines use the nonlinear equations; dashed lines use the upright linearization. Matching colors indicate matching forces.

Table 3: Percentage model errors under constant cart force, relative to the nonlinear response at each comparison time.

Force (N)	0.1 s		0.25 s		0.5 s	
	e_x (%)	e_θ (%)	e_x (%)	e_θ (%)	e_x (%)	e_θ (%)
0.5	8.02×10^{-6}	1.33×10^{-5}	0.0004	0.0006	0.0096	0.0133
1	3.21×10^{-5}	5.31×10^{-5}	0.0014	0.0023	0.0385	0.0531
2	0.0001	0.0002	0.0058	0.0092	0.1534	0.2113
4	0.0005	0.0008	0.0232	0.0368	0.6023	0.8302
8	0.0021	0.0034	0.0923	0.1466	2.2531	3.1122
16	0.0082	0.0136	0.3645	0.5794	7.3477	10.259

At 0.5 s, the 0.5 N input gives an angle error of 0.013%, while 16 N gives 10.26%. Stronger forces drive the pendulum farther from upright, increasing the effect of the nonlinear terms. Together, the release and force comparisons support using the linear model for local controller design. They also establish the need to test that controller on the nonlinear model, as done in Appendices C and D.

Appendix B. Controller selection, LQR design, and verification

With the controllable linear model from Appendix A, we next choose how to generate the cart force. A proportional-integral-derivative (PID) controller combines the current output error, its accumulated history, and its rate of change. Its gains can be tuned from measured responses or with a model [4]. For this plant, controlling angle and cart position with PID would require coordinating the two objectives, for example through nested loops.

Pole-placement feedback also uses a model and the system states. The designer selects the desired closed-loop poles, and MATLAB calculates a gain that produces them [5]. This directly specifies response modes but does not explicitly assign a cost to actuator effort. A linear quadratic regulator (LQR) instead calculates state feedback from penalties on state deviations and force. Model predictive control (MPC) repeatedly predicts future motion and optimizes a sequence of inputs while accounting for constraints [6]. It can represent force and track limits directly, but requires an optimization at each control update.

We select LQR because the model and all four simulated states are available. A single feedback law can address pendulum balance and cart motion together, and its cost function provides a direct way to trade motion against force demand. The resulting gain is calculated once and applied through a matrix multiplication. Unlike constrained MPC, this LQR design does not enforce the actuator limit; we examine that limit separately below.

To calculate the gain, we minimize the accumulated quadratic cost,

$$J = \int_0^\infty (z^T Q z + R u^2) dt = \int_0^\infty (4x^2 + 2\dot{x}^2 + 300\theta^2 + 10\dot{\theta}^2 + 0.5u^2) dt.$$

Here x is measured in meters, θ in radians, and u in newtons. The four diagonal entries of Q penalize cart position, cart velocity, pendulum angle, and angular velocity in that order. The scalar R penalizes force. Increasing a weight makes the corresponding squared quantity more expensive in the optimization. Since the states have different units, the coefficients must be interpreted together with those units; their numerical sizes alone do not establish relative importance.

The baseline settings in Table 4 are tuning choices. MATLAB’s `lqr` function solves the algebraic Riccati equation and returns the gain K and the poles of $A - BK$ [3]. The control law is $u = -Kz$. All four poles have negative real parts, establishing stability of the unlimited linear closed loop. The following tests examine its response and assess the weight choices.

Table 4: Baseline LQR settings, feedback gain, and closed-loop poles.

Quantity	Baseline value
Q	$\text{diag}(4, 2, 300, 10)$
R	0.5
K	$(-2.8284, -6.3204, -86.8641, -25.2346)$
Slow pole pair	$-0.699 \pm 0.548i$
Fast pole pair	$-4.042 \pm 1.125i$

First, we compare the uncontrolled and baseline LQR-controlled linear plants from the same 5° release, with the cart at the origin and both velocities zero. Figure 6 shows the first second. The uncontrolled angle grows away from upright, while feedback reverses that growth. The cart initially moves to correct the pendulum angle; keeping the cart stationary throughout this transient would prevent that corrective motion.

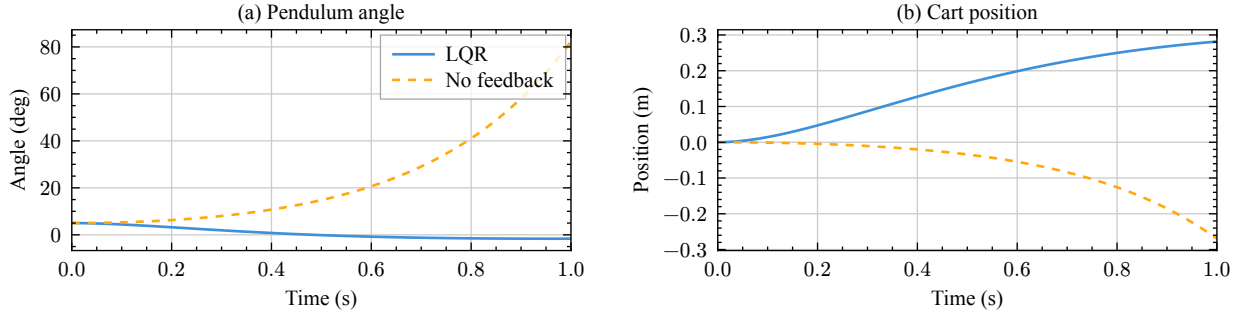


Figure 6: Effect of LQR feedback during the first second of a 5° release on the linear plant.

The longer response in Figure 7 shows the cart returning toward the origin as the pendulum settles. These simulations use ode45 with relative tolerance 10^{-10} , absolute tolerance 10^{-12} , and a maximum step of 0.005 s. Results are evaluated every 0.01 s over 30 s, with every second sample used for plotting. Unlimited responses are checked against the matrix-exponential solution at the final time.

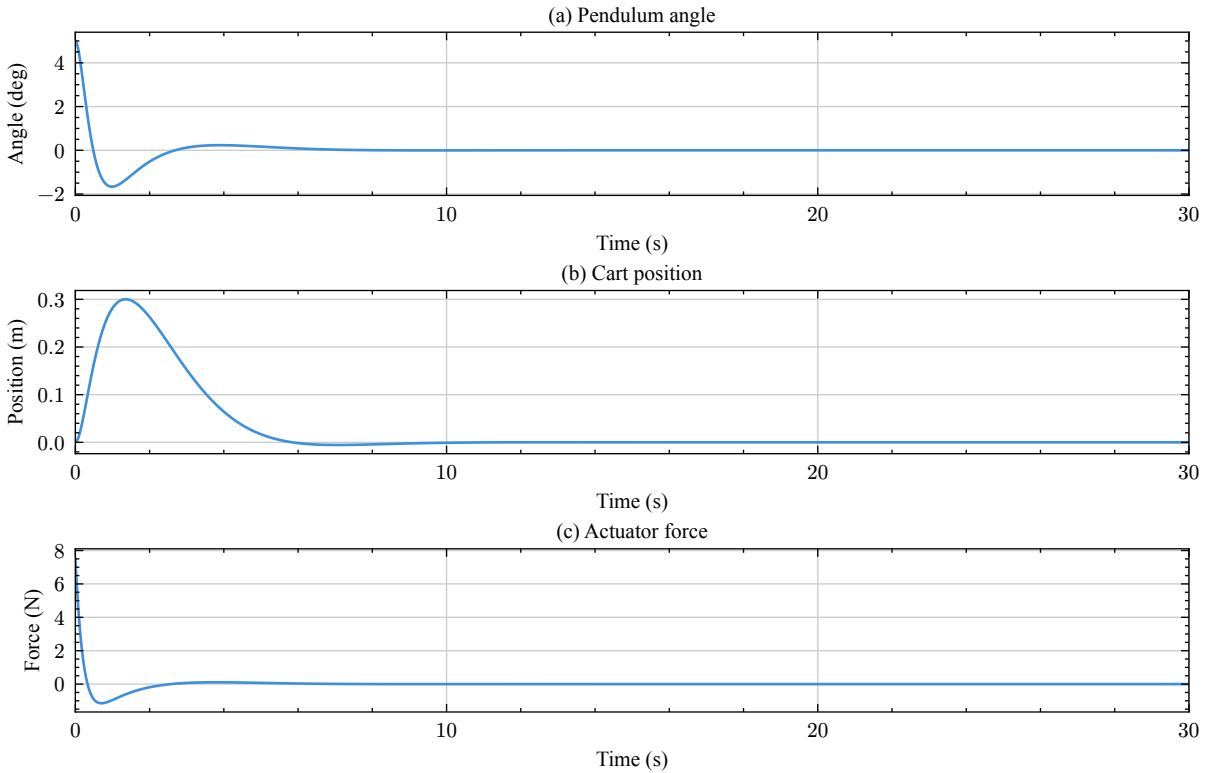


Figure 7: Baseline unlimited LQR response over 30 s on the linear plant, starting from 5° .

For this appendix, joint settling time t_s is the first sampled time after which both $|\theta| \leq 1^\circ$ and $|x| \leq 0.02$ m hold through the end of the run. The baseline meets these conditions at 4.91 s, with peak cart displacement 0.3 m and peak force 7.58 N. Its final state norm is 6.03×10^{-10} , consistent with convergence to equilibrium.

Having verified the baseline response, we vary one penalty at a time to assess the tuning tradeoffs. All runs use the same linear plant, unlimited actuator, and 5° release. In Figure 8, the angle penalty q_θ takes values of 100, 300, and 900, while the remaining state weights and $R = 0.5$ stay fixed. The response details show the first 10 s; the metrics use all 30 s.

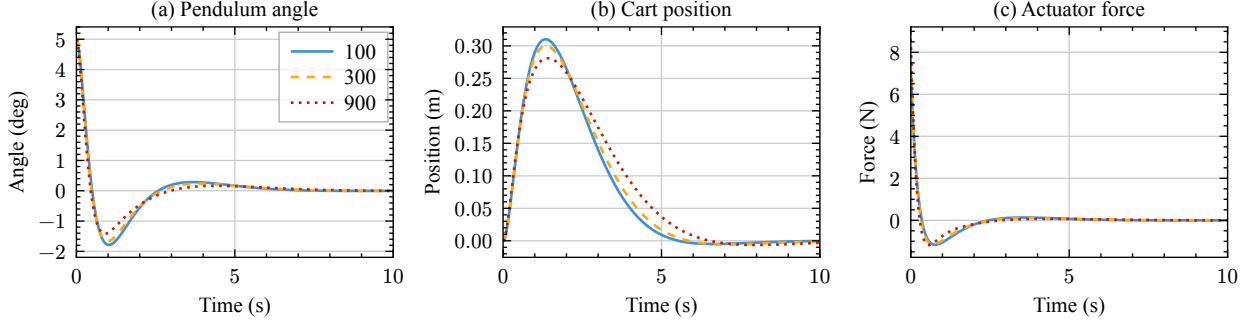


Figure 8: Effect of changing the angle penalty q_θ from the baseline value of 300. The legend gives q_θ ; all other penalties remain fixed.

We next keep $Q = \text{diag}(4, 2, 300, 10)$ fixed and vary the force penalty R over 0.1, 0.5, and 2. Figure 9 shows how this changes the motion and actuator demand.

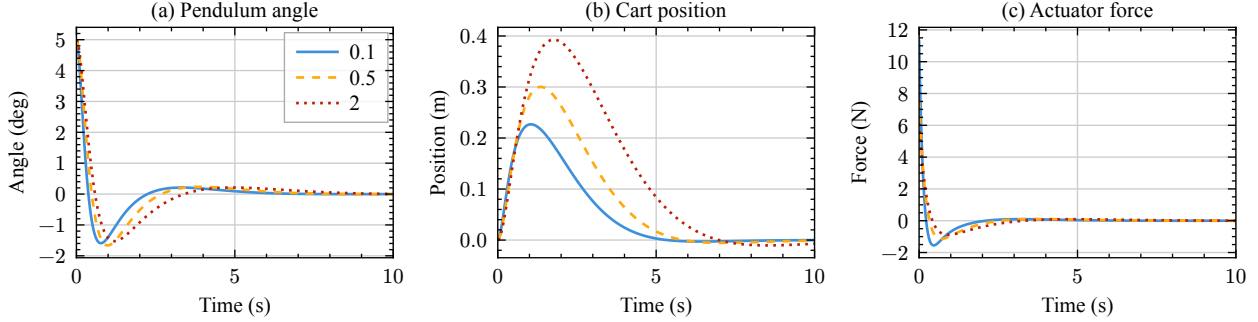


Figure 9: Effect of changing the force penalty R from the baseline value of 0.5. The legend gives R ; all state penalties remain fixed.

To quantify these differences, Table 5 reports the joint settling time t_s , peak cart displacement x_{peak} , peak requested force u_{peak} , and force effort $E_u = \int_0^{30} u(t)^2 dt$. The baseline appears once because it belongs to both sweeps.

Table 5: Weight-sweep metrics for the unlimited linear plant following a 5° release.

q_θ	R	t_s (s)	x_{peak} (m)	u_{peak} (N)	E_u (N ² s)
100	0.5	4.64	0.31	7.16	4.99
300	0.5	4.91	0.3	7.58	5.09
900	0.5	5.49	0.281	8.62	5.49
300	0.1	4.14	0.227	11.38	7.42
300	2	6.2	0.393	6.13	4.16

Increasing q_θ from 100 to 900 reduces peak cart travel from 0.310 to 0.281 m, but raises peak force from 7.16 to 8.62 N and joint settling time from 4.64 to 5.49 s. Reducing R to 0.1 gives faster settling and less travel, but its 11.38 N peak exceeds the intended actuator limit. Increasing R to 2 reduces force effort but increases travel and settling time. The baseline retains a peak force below 10 N for this release while recovering faster and using less travel than $R = 2$. We therefore retain it as a compromise for the nonlinear tests.

The preceding LQR calculations assume that the actuator can deliver every requested force. We now isolate the effect of its 10 N limit by using the same linear plant and gain with either $u = -Kz$ or $u = \text{sat}(-Kz)$, where saturation clips the command to $[-10, 10]$ N. We test 5° and 20° releases without disturbance. The larger release activates the force limit. Figure 10 compares the angle and cart responses.

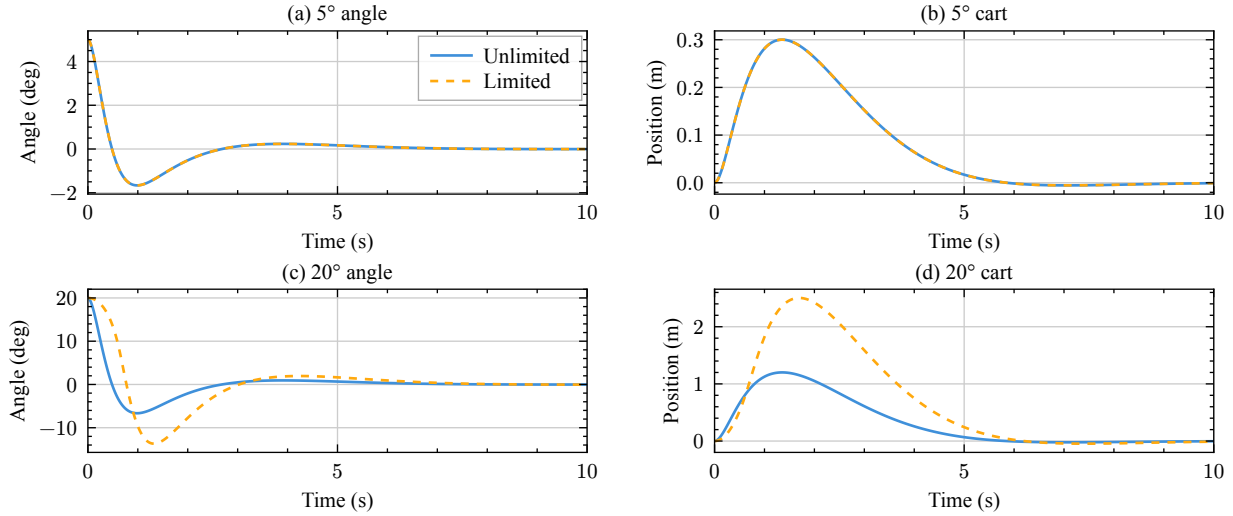


Figure 10: Linear-plant responses with unlimited (solid) and 10 N limited (dashed) feedback.

To explain these response differences, Figure 11 compares the requested and applied forces. The 20° case shows the applied force clipped while the requested command exceeds 10 N.

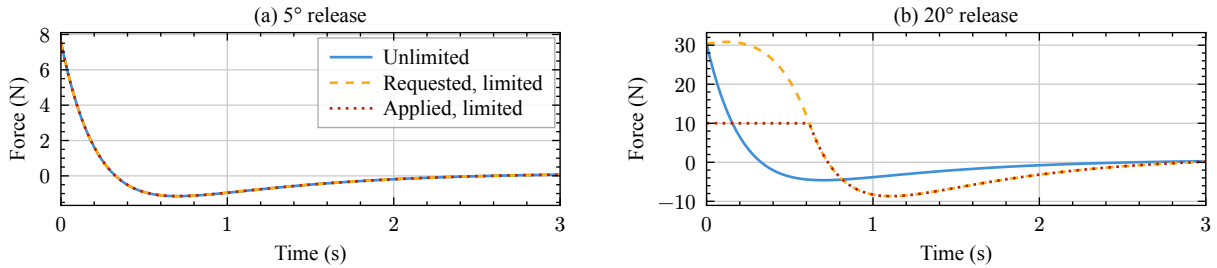


Figure 11: Unlimited force, limited-loop requested force, and limited-loop applied force over the first 3 s. Applied force is clipped at ± 10 N.

Table 6 summarizes these runs. Recovery requires the angle and cart position to remain within the settling bounds during the final 2 s. Saturation duration is the time for which the requested command exceeds 10 N in magnitude, estimated on the 0.01 s analysis grid.

Table 6: Linear-plant recovery and actuator saturation with the baseline gain. Peak force is the requested command.

Release (deg)	Actuator	Recovered	t_s (s)	Saturated (s)	u_{peak} (N)
5	Unlimited	Yes	4.91	0	7.58
5	10 N limit	Yes	4.91	0	7.58
20	Unlimited	Yes	7.61	0	30.32
20	10 N limit	Yes	9.28	0.62	30.86

At 5° , the requested force remains below 10 N, so the unlimited and limited responses coincide. At 20° , the limit increases settling time from 7.61 to 9.28 s and peak cart travel from 1.201 to 2.504 m. The actuator saturates for approximately 0.62 s, but the system still recovers. Once clipping occurs, the dynamics are no longer governed by the fixed matrix $A - BK$ alone. Its stable poles therefore do not guarantee recovery for arbitrary releases. Appendix D tests the limited controller on the nonlinear plant.

The essential controller calculation is shown below. It constructs the linear model, sets the penalties, and calculates the feedback gain. The separate `controller_studies.m` file runs the response, weight, and saturation comparisons and exports the data used in this appendix.

```
function K = design_controller(M, m, ell, c_theta, g)
% State order is [x; x_dot; theta; theta_dot].
A = [0 1 0 0;
     0 0 -m*g/M c_theta/(M*ell);
     0 0 0 1;
     0 0 (M+m)*g/(M*ell) -c_theta*(M+m)/(M*m*ell^2)];
B = [0; 1/M; 0; -1/(M*ell)];
Q = diag([4 2 300 10]);
R = 0.5;
K = lqr(A, B, Q, R);
end
```

To apply the same gain to the original nonlinear plant, the Simulink block uses the acceleration function below. Its eighth input is the total cart force after actuator limiting and disturbance addition.

```
function [theta_ddot, x_ddot] = accelerations(theta_dot, theta, M, m, ell, c_theta, g, F_cart)
% Nonlinear cart-pendulum equations with an applied horizontal cart force.
% Positive theta is the upright perturbation used in Mini Project 1.

s = sin(theta);
c = cos(theta);
mass_matrix = [M + m, m * ell * c; m * ell * c, m * ell^2];
forcing = [F_cart + m * ell * s * theta_dot^2; ...
          m * g * ell * s - c_theta * theta_dot];
acceleration = mass_matrix \ forcing;

x_ddot = acceleration(1);
theta_ddot = acceleration(2);
end
```

Appendix E shows how these functions connect through the feedback loop. With the gain fixed, Appendix C defines the nonlinear release and random-force test protocol, and Appendix D reports the resulting operating limits.

Appendix C. Random-force generation, shaping, and test protocol

With the controller fixed in Appendix B, we first test it using random forces that change every 0.1 s. Each new value is independent of the previous one, so the force can jump from near -10 N to near $+10$ N at a sample boundary. Such sudden jumps are useful for stressing the controller, but we also want to test a force that changes gradually, as a physical disturbance acting on a cart might. We therefore smooth the random force with Gaussian averaging and limit how quickly it can change.

To generate the original force, MATLAB draws independent uniform values ξ_k in $[-1, 1]$. Multiplying by the force bound a gives $r(t) = a\xi_k$, held until the next draw. We use bounds of 2.5, 5, 7.5, and 10 N. The same seeded draws are scaled for each bound, so their signs and timing match.

Gaussian averaging replaces each force value with a weighted average of nearby values. Samples closest in time receive the most weight. For the smoothed force s_i , we use

$$w_j = \exp\left(-\frac{(jh)^2}{2\sigma^2}\right), \quad s_i = \frac{\sum_{j=-m}^m w_j r_{i+j}}{\sum_{j=-m}^m w_j}, \quad m = \left\lceil 3\frac{\sigma}{h} \right\rceil.$$

Here σ controls how much smoothing occurs. A larger value blends a longer stretch of the force record and produces gentler changes. We calculate the average every $h = 0.002$ s, using samples up to 3σ before and after each point. At the ends, we extend the nearest force value. Because the average uses future samples, we generate the complete record before simulation.

Smoothing alone can still leave a rapid change. We therefore add a rate limiter [7] that allows the force to move only a fixed amount per time step,

$$d_{i+1} = d_i + \text{clip}(s_{i+1} - d_i, -Lh, Lh).$$

The function clip restricts each change to $\pm Lh$, where L is the allowed rate in N/s. We choose $\sigma = 0.1$ s and $L = 50$ N/s. The limiter starts at zero for random trials and at -10 N for the reversal test.

We first check these operations with one clear example. In Figure 12, the target switches from -10 to $+10$ N at 0.5 s. We compare the original jump, Gaussian averaging, rate limiting, and both operations together.

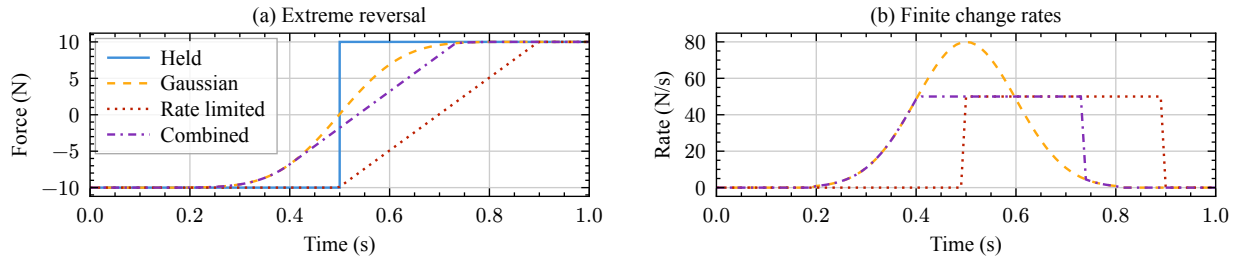


Figure 12: Force reversal at 0.5 s and the finite change rates of the three continuous profiles. The held force jumps instantaneously and has no finite rate at that instant.

The Gaussian curve rounds the jump, while the limiter caps its slope at 50 N/s. A full 20 N change at that rate needs at least 0.4 s. Table 7 measures time from the target jump to $+9$ N. Centered averaging begins changing earlier, so these times are not full transition durations.

Table 7: Reversal timing and force-change rates with $\sigma = 0.1$ s and $L = 50$ N/s. A dash denotes an undefined instantaneous rate.

Profile	Maximum rate (N/s)	Time after jump to +9 N (s)
Held	—	0
Gaussian	79.98	0.16
Rate limited	50	0.38
Combined	50	0.22

With the single jump explained, Figure 13 shows the same operations on the random-force sequence from seed 79202. Panel (a) shows the first 10 s, and panel (b) enlarges the first 3 s. The original force has sharp steps; Gaussian averaging rounds them, and the combined curve also obeys the rate limit.

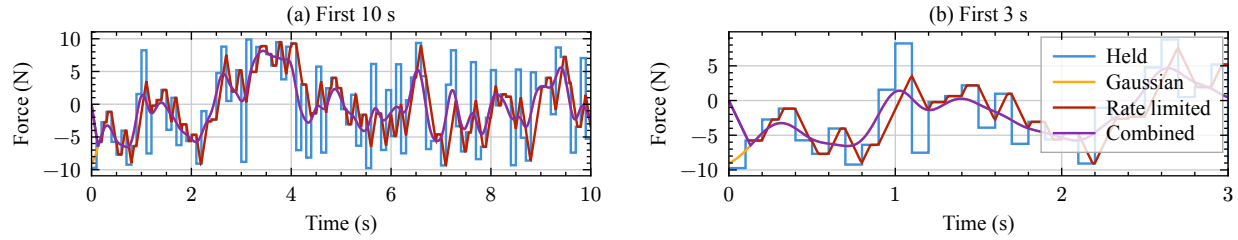


Figure 13: Held, Gaussian-averaged, rate-limited, and combined disturbances from the same 10 N target sequence, seed 79202.

The curves stay within the original force bounds. However, smoothing removes short peaks and reduces root-mean-square (RMS) force, as Table 8 shows for the full 30 s record. Thus, the shaped force is smoother and also weaker. In Simulink, we interpolate between its fine-grid samples; the original force remains held between draws.

Table 8: Actual strength and slope of the four 30 s disturbance profiles.

Profile	Peak (N)	RMS (N)	Maximum rate (N/s)
Held	9.99	5.95	—
Gaussian	9.03	3.23	60.51
Rate limited	9.58	4.11	50
Combined	8.19	3.2	50

We next check how the settings change the force. Figure 14 compares Gaussian widths of 0.05, 0.1, and 0.2 s without a limiter, then rate limits of 25, 50, and 100 N/s without averaging. Using the same 10 N target makes the effect of each setting visible.

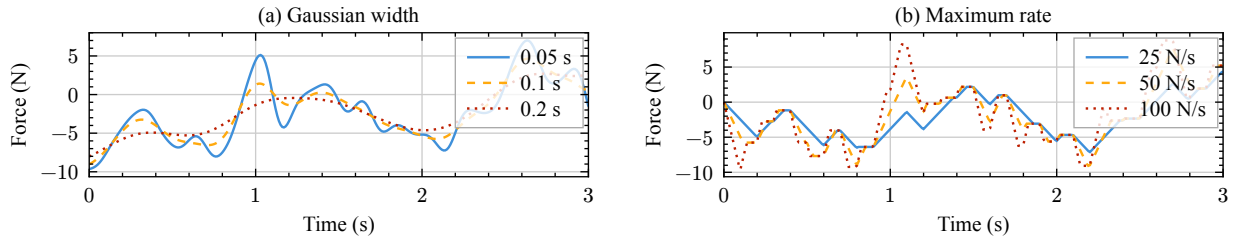


Figure 14: Sensitivity of the disturbance to Gaussian width and rate limit, tested separately.

Larger Gaussian widths blend away more short fluctuations. Lower rate limits produce slower ramps. Table 9 quantifies these changes over 30 s. We retain $\sigma = 0.1$ s and $L = 50$ N/s for the controller comparisons.

Table 9: Separate shaping sweeps using the same 10 N random target.

Operation	Setting	Peak (N)	RMS (N)	Maximum rate (N/s)
Gaussian	0.05 s	9.64	4.14	146.32
Gaussian	0.1 s	9.03	3.23	60.51
Gaussian	0.2 s	8.03	2.57	22.06
Limiter	25 N/s	9.27	3.49	25
Limiter	50 N/s	9.58	4.11	50
Limiter	100 N/s	9.94	4.68	100

Finally, 160 nonlinear trials compare held and combined disturbances across seeds 79202–79221 and four bounds. Each starts upright at rest and lasts 30 s. The ode45 settings are a 0.005 s maximum step, 10^{-7} relative tolerance, and 10^{-9} absolute tolerance. States are sampled every 0.02 s; balance requires $|\theta| < 90^\circ$ throughout. Appendix D reports angle RMS, peak travel, saturation, and paired-seed results. Gaussian averaging changes temporal correlation and RMS strength, so these tests assess both effects together.

Appendix D. Nonlinear controller performance and operating limits

Appendix B gives us a controller designed around the upright linear model, and Appendix C defines the forces used to test it. Here we apply that controller to the nonlinear Simulink plant and ask four questions. How far can the pendulum start from upright and still recover? What changes when it fails? How does Gaussian width affect the response? Does smoothing still change the response when the forces have equal RMS strength? The gain and actuator limit remain fixed throughout.

We first revisit the original release tests in Table 10. Each starts with the cart and pendulum at rest, with no disturbance. The 20° release recovers, while the 30° release fails. Recovery is assessed over 30 s: the final angle and final-two-second angle RMS must be below 2° . In the new tests, the pendulum must also stay within $\pm 90^\circ$ throughout.

Table 10: Original release tests. Failed-case travel includes motion after loss of balance on the model’s unlimited track.

Initial angle (deg)	Peak cart displacement (m)	Final 2 s RMS (deg)	Saturation (%)	Recovered
5	0.301	0	0	Yes
10	0.646	0	0.33	Yes
20	2.977	0	3.93	Yes
30	1799.904	227.543	100	No
45	1800.317	203.3561	100	No
60	1800.069	217.6301	100	No

To locate the transition more closely, we test every degree from 20° to 30° , then test the intervening interval in 0.1° increments. Figure 15 shows the travel and saturation time for each release. Failed-case metrics stop at the first sampled 90° crossing, so falling motion does not dominate the vertical scale. Successful-case metrics use the full 30 s.

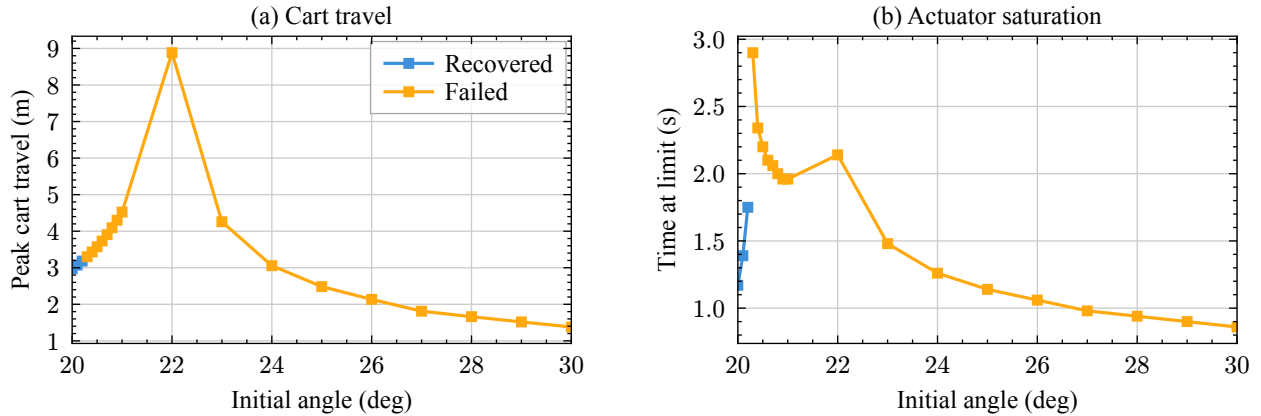


Figure 15: Refined release sweep. Metrics for failed trials cover only the motion up to loss of balance.

The largest tested release that recovers is 20.2° the next release, 20.3° , loses balance. The recovery-to-failure boundary therefore lies between these two tested angles for the stated initial conditions. Recovery at 20.2° still requires 3.18 m of peak cart travel and 1.75 s at the force limit, so balancing alone does not establish feasibility on a short track.

All new trials use ode45 with a 0.005 s maximum step, 10^{-7} relative tolerance, and 10^{-9} absolute tolerance. Metrics use a 0.02 s state grid. These tests hold the controller and plant parameters fixed, so their conclusions describe the tested releases and disturbance profiles for this model.

The refined sweep identifies two neighboring releases with different outcomes. We now compare their histories in Figure 16 to see how those outcomes develop. Both use the same controller and 10 N actuator limit; only the starting angle changes. The failed trace ends when the pendulum first reaches 90° .

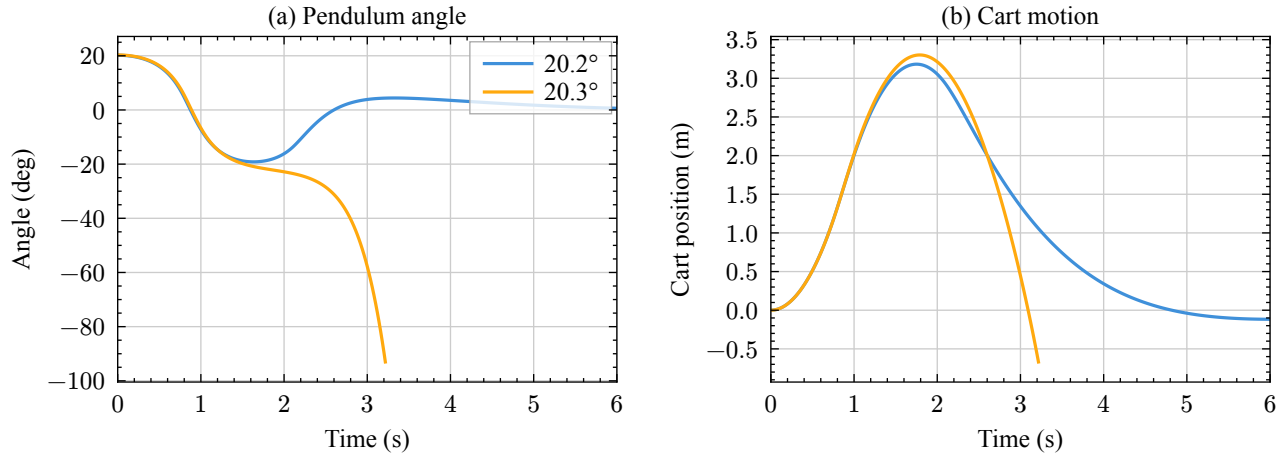


Figure 16: Near-boundary releases. One recovers; the next tested release loses balance.

The cart must move to bring the pendulum back toward upright. That motion is governed by the force the actuator can deliver. The controller may request more than 10 N, but the plant receives only the clipped value. The following comparison shows the requested and applied forces separately for each release.

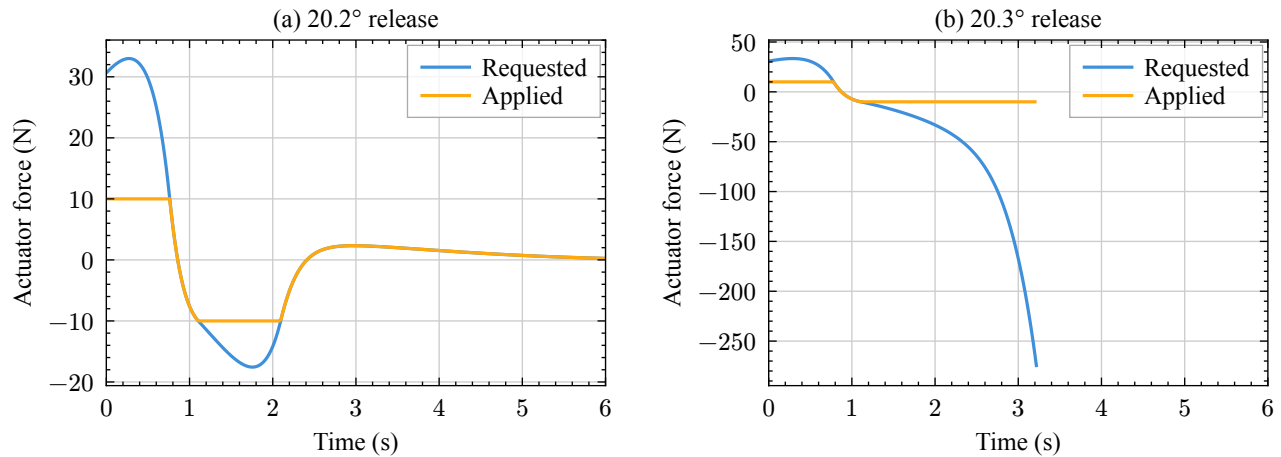


Figure 17: Requested and applied force near the recovery boundary. The applied force is limited to ± 10 N.

For the 20.2° release, the controller eventually brings its requested force back within the actuator range and the pendulum recovers. At 20.3° , the pendulum reaches 90° after 3.22 s. By that point, the requested force reaches approximately 276 N in magnitude, while the applied force remains limited to 10 N. The large gap shows why the actuator cannot deliver the correction requested by this gain. Saturation also occurs in the successful trial; its presence alone does not determine the outcome.

Having examined releases, we next start the system upright and apply random forces. Figure 18 compares the original held force with Gaussian averaging followed by rate limiting. Both use the same 5 N target and seed 79202. The shaped version uses $\sigma = 0.1$ s and $L = 50$ N/s. Every disturbance trial lasts 30 s; balance requires $|\theta| < 90^\circ$ throughout.

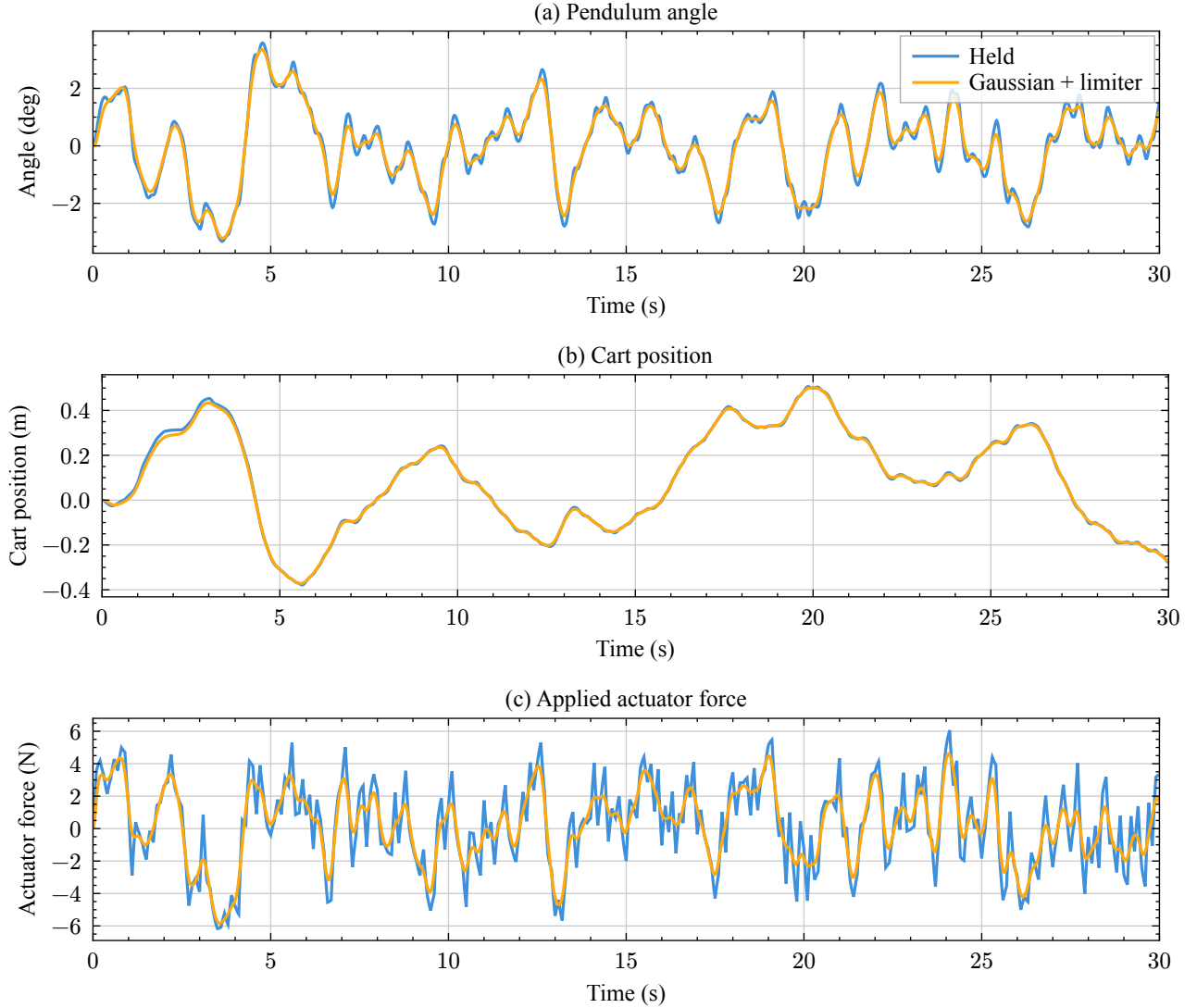


Figure 18: Nonlinear responses to held and shaped 5 N disturbances, seed 79202.

At this bound, both trials maintain balance. Shaping reduces force RMS from 2.98 to 1.61 N, while angle RMS falls from 1.39 to 1.31° . The cart travels nearly the same distance. Table 11 extends this comparison to the other bounds. Its large failed-case values include motion after the pendulum falls.

Table 11: Held and shaped disturbance results for seed 79202.

Bound (N)	Profile	Balanced	Force RMS (N)	Angle RMS (deg)	Peak cart (m)	Saturated (%)
2.5	Held	Yes	1.49	0.69	0.253	0
2.5	Shaped	Yes	0.81	0.66	0.251	0
5	Held	Yes	2.98	1.39	0.506	0
5	Shaped	Yes	1.61	1.31	0.502	0
7.5	Held	Yes	4.46	2.08	0.759	0
7.5	Shaped	Yes	2.4	1.96	0.752	0
10	Held	No	5.95	2371.14	1455.883	88.74
10	Shaped	No	3.2	492.06	1069.093	87.67

A single random sequence does not show how often a given bound causes failure. We therefore repeat the held-versus-shaped comparison for seeds 79202–79221. Figure 19 shows the number of trials that maintain balance and their median angle RMS. This separates the balance outcome from the amount of motion in successful trials.

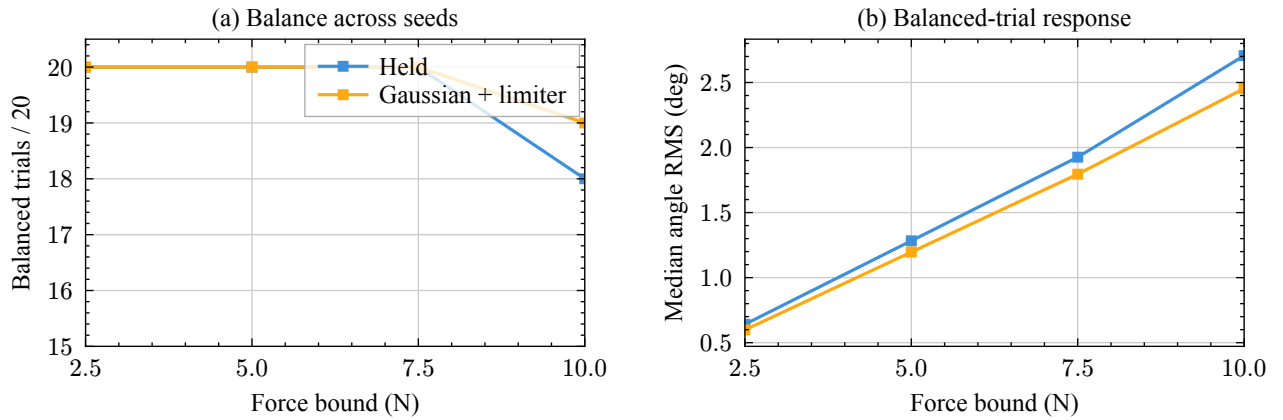


Figure 19: Twenty-seed comparison of held and shaped disturbances. Angle medians include balanced trials only.

Both profiles maintain balance in every tested seed through 7.5 N. At 10 N, the held version succeeds in 18 of 20 trials and the shaped version in 19 of 20. Seed 79202 fails with both versions. Table 12 also gives the actual force strength and cart travel, which help interpret the response differences.

Table 12: Results across seeds 79202–79221. Response medians include balanced trials only.

Bound (N)	Profile	Balanced / 20	Median force RMS (N)	Median angle RMS (deg)	Median peak cart (m)
2.5	Held	20	1.44	0.64	0.209
2.5	Shaped	20	0.77	0.6	0.207
5	Held	20	2.87	1.28	0.417
5	Shaped	20	1.55	1.2	0.414
7.5	Held	20	4.31	1.93	0.626
7.5	Shaped	20	2.32	1.8	0.621
10	Held	18	5.75	2.71	0.87
10	Shaped	19	3.09	2.45	0.817

The nominal velocity histories below complete the four-state record used by the controller. After a release, the velocities decay toward rest. Under an ongoing disturbance, they continue to fluctuate even when the pendulum remains balanced.

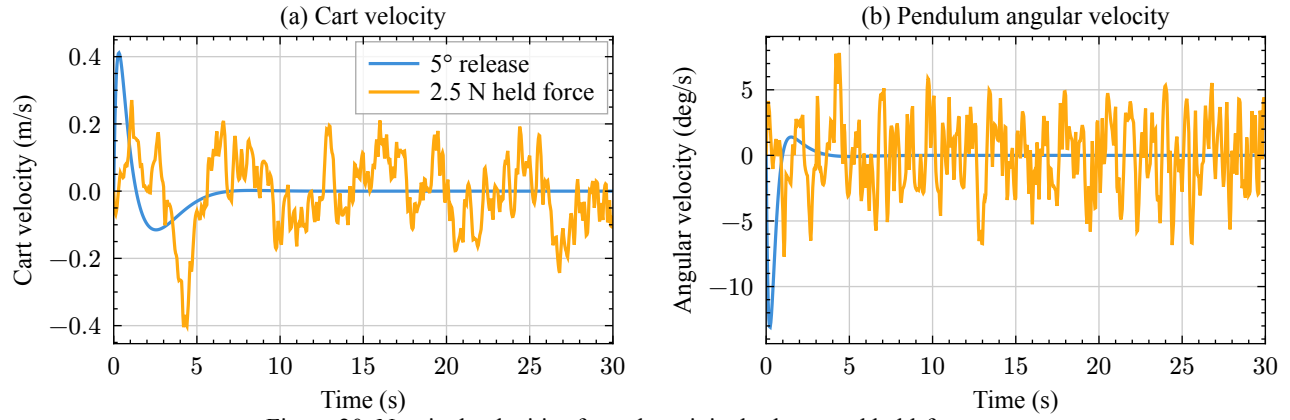


Figure 20: Nominal velocities from the original release and held-force tests.

The corresponding original held-force actuator histories below show how increasing the force bound changes demand. At 10 N for seed 79202, loss of balance leads to sustained actuator saturation. The successful lower-bound trials continue correcting the random force without remaining at the limit.

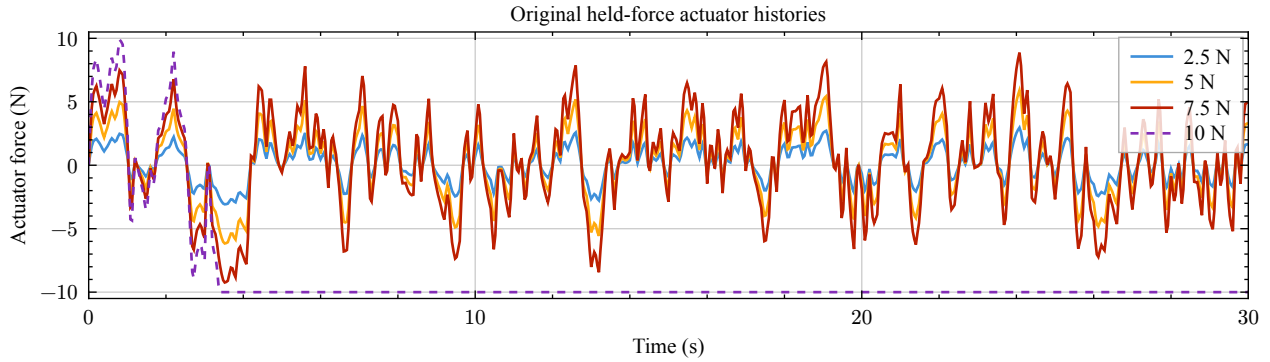


Figure 21: Original actuator histories for the four held-force bounds. The failed 10 N case is dashed.

These velocity and force histories support the balance metrics above. They also show that remaining upright under a continuing disturbance does not mean the cart and pendulum have come to rest.

The preceding comparison uses one Gaussian width. Appendix C shows that changing this width changes the force waveform; here we test how it changes the controller's response. For each σ of 0.05, 0.1, and 0.2 s, we run all four force bounds and all 20 seeds with the same 50 N/s rate limit. This gives 240 nonlinear trials, including a repeated baseline width for direct consistency checks.

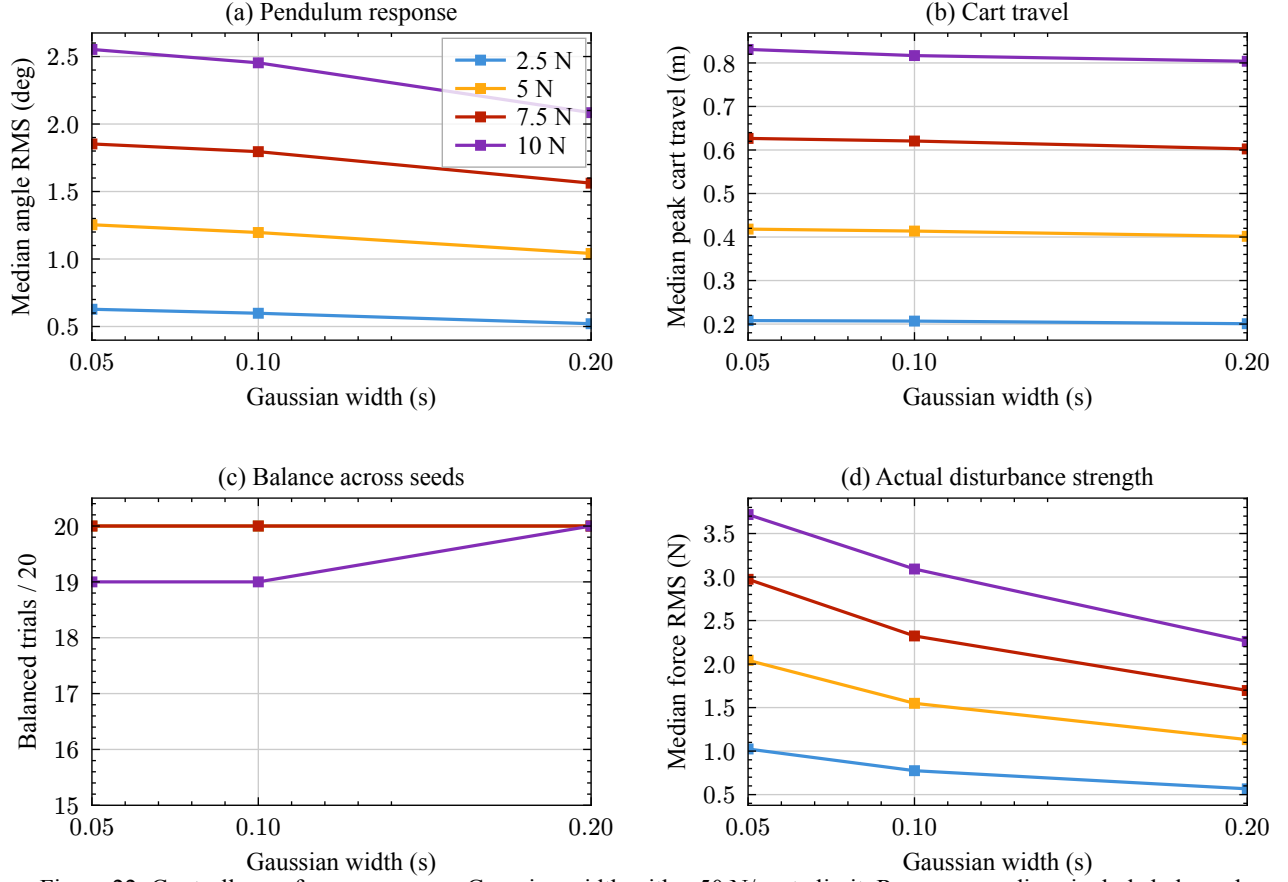


Figure 22: Controller performance versus Gaussian width with a 50 N/s rate limit. Response medians include balanced trials only.

Increasing the Gaussian width reduces median angle motion and cart travel at every tested bound. At 10 N, widths of 0.05 and 0.1 s maintain balance in 19 of 20 seeds, while 0.2 s maintains balance in all 20. However, median force RMS also falls from 3.72 to 2.26 N across that sweep. These results describe the combined effect of gentler variation and lower force strength. The next comparison holds RMS strength fixed.

Table 13 collects the results behind the width comparison. All lower-bound trials remain balanced at every tested width. The differences are most visible at 10 N, where the widest average removes enough rapid variation and force strength to keep all tested seeds balanced. Angle and travel medians use successful trials only; the balance count separately records failures.

Table 13: Gaussian-width results across the same 20 seeds at each force bound.

Width (s)	Bound (N)	Balanced / 20	Median force RMS (N)	Median angle RMS (deg)	Median peak cart (m)
0.05	2.5	20	1.02	0.63	0.208
0.05	5	20	2.04	1.25	0.418
0.05	7.5	20	2.97	1.85	0.627
0.05	10	19	3.72	2.55	0.831
0.1	2.5	20	0.77	0.6	0.207
0.1	5	20	1.55	1.2	0.414
0.1	7.5	20	2.32	1.8	0.621
0.1	10	19	3.09	2.45	0.817
0.2	2.5	20	0.57	0.52	0.201
0.2	5	20	1.13	1.04	0.402
0.2	7.5	20	1.7	1.56	0.602
0.2	10	20	2.26	2.08	0.804

The width comparison motivates one final test. Because wider averaging reduces force strength, the preceding tests combine two effects. We now separate them by making the held and shaped forces have the same 2 N RMS over 30 s. For each seed, we generate both versions from the same 10 N target, then scale each downward to the common RMS level. Downward scaling preserves the 10 N amplitude bound and the shaped profile's 50 N/s rate limit. It also avoids clipping after scaling, which would change the waveform.

For these equal-RMS trials, the largest applied disturbance peaks are 3.59 N for held inputs and 6.06 N for shaped inputs. The shaped-force rate never exceeds 38.14 N/s. The force history below illustrates how equal overall RMS can coexist with different peaks and timing.

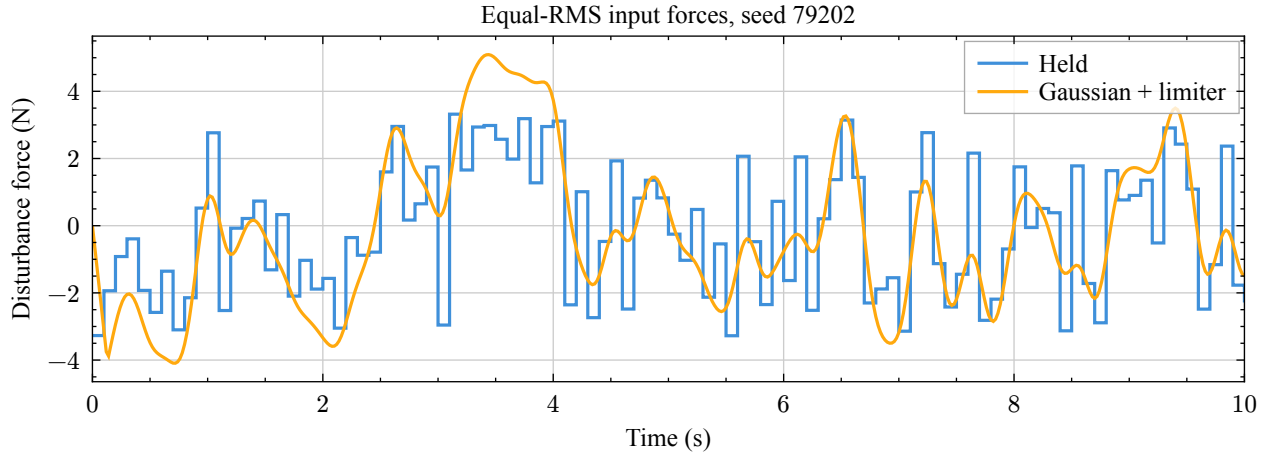


Figure 23: Held and shaped forces scaled to 2 N RMS over 30 s. The first 10 s are shown.

With the input strength matched, Figure 24 compares the resulting angle motion and cart travel for each seed. The dashed line marks equal response; points above it indicate greater motion under the shaped force.

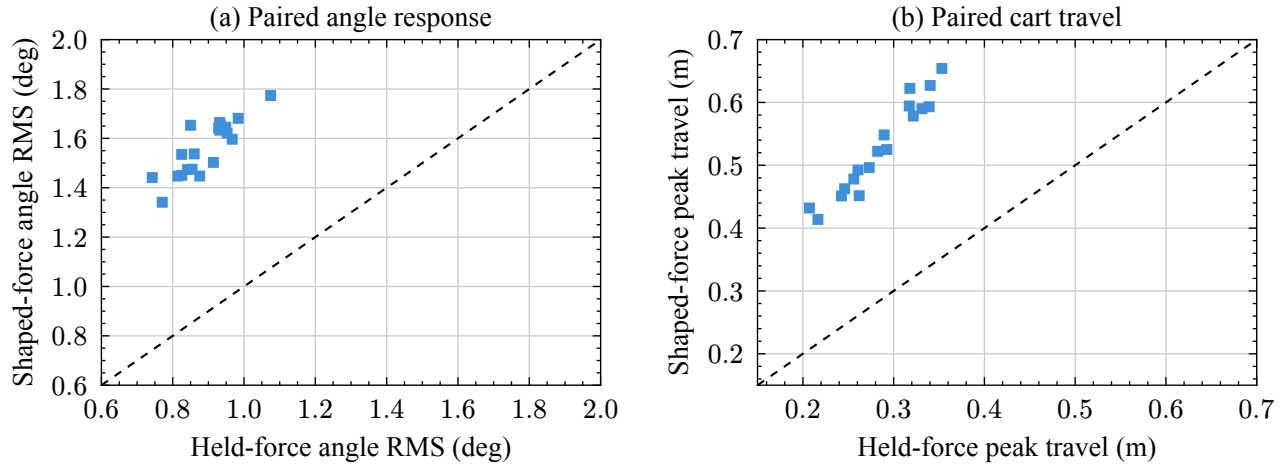


Figure 24: Equal-RMS comparisons. Each point pairs the same seed; points above the dashed equality line indicate a larger response to the shaped force.

At equal RMS strength, the shaped force produces a larger response. Median angle RMS increases from 0.89 to 1.57°, and median peak cart travel increases from 0.291 to 0.537 m. All 20 seeds remain balanced with both profiles, and no trial reaches the actuator limit. The shaped force varies over longer intervals and has different peaks, so equal RMS does not imply equal effect on the cart and pendulum. Gaussian smoothing therefore cannot be described as universally improving controller performance.

Table 14: Equal-RMS results at 2 N across seeds 79202–79221.

Profile	Balanced / 20	Median force RMS (N)	Median angle RMS (deg)	Median peak cart (m)	Median saturation (%)
Held	20	2	0.89	0.291	0
Shaped	20	2	1.57	0.537	0

To make the paired statistics easier to interpret, the following histories show seed 79202 at the same RMS strength. They reveal when the shaped force produces a different angle or actuator response, even though its overall RMS is equal.

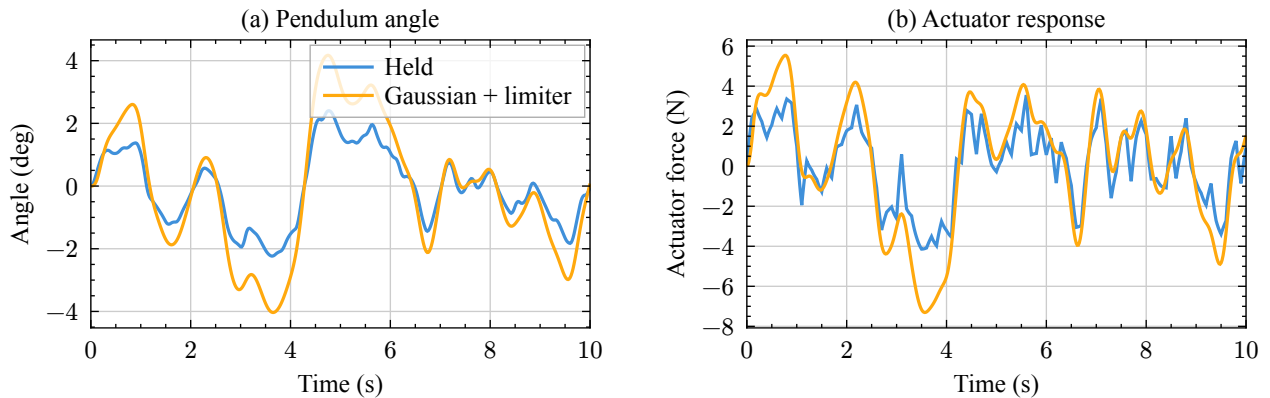


Figure 25: Example equal-RMS controller responses for seed 79202, showing the first 10 s.

Appendix E. Simulink feedback loop and nonlinear plant

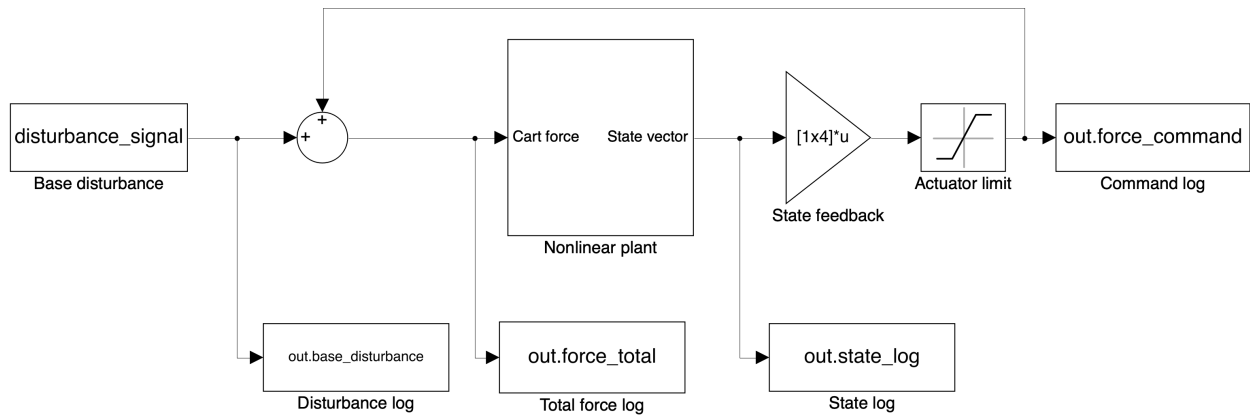


Figure 26: Complete feedback loop with state feedback, actuator saturation, an external disturbance, and signal logging.

The gain from Appendix B acts on all four simulated states. The actuator block limits the command, then the sum adds the disturbance before the total force reaches the cart. Output blocks record the states, command, disturbance, and total force. This wiring implements $F = \text{sat}(-Kz) + d$.

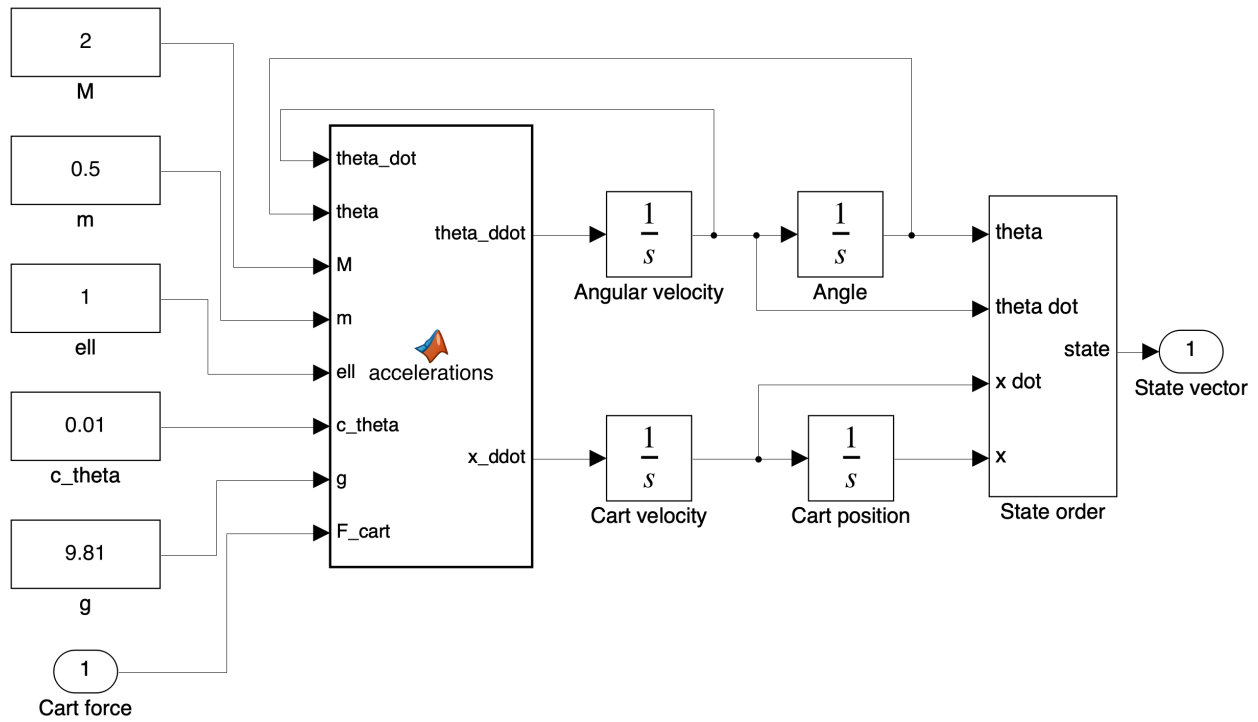


Figure 27: Nonlinear equations and four integrators retained from Project 1.

Inside the preceding feedback loop, the acceleration function from Appendix B drives two integrator chains. The State order block receives $(\theta, \dot{\theta}, \dot{x}, x)$ from top to bottom and rearranges them internally into $(x, \dot{x}, \theta, \dot{\theta})$ for the gain. The display order therefore follows the diagram, while the vector sent to the controller follows the design matrices.

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